

AI3602 Data Mining Homework 1.1

Kailing Wang 521030910356

March 11.2024

Question 1

A metric d satisfies:

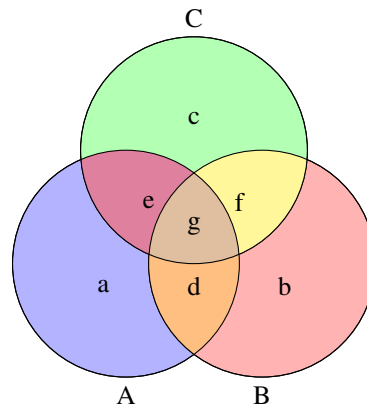
- 1) $d(i, j) \geq 0$
- 2) $d(i, i) = 0$
- 3) $d(i, j) = d(j, i)$
- 4) $d(i, j) \leq d(i, k) + d(k, j)$.

a. The Jaccard distance is: $J(A, B) = 1 - \frac{|A \cap B|}{|A \cup B|}$. Suppose A or B is not empty, which is trivial.

- 1) $|A \cap B| \leq |A \cup B| \neq 0, \frac{|A \cap B|}{|A \cup B|} \leq 1: J(A, B) \geq 0$.
- 2) $\forall A, |A \cap A| = |A \cup A| = |A|, J(A, B) = 1 - 1 = 0$.
- 3) $|A \cap B| = |B \cap A|, |A \cup B| = |B \cup A|,$
 $J(A, B) = 1 - \frac{|A \cap B|}{|A \cup B|} = 1 - \frac{|B \cap A|}{|B \cup A|} = J(B, A)$.
- 4) To prove $J(A, B) \leq J(A, C) + J(C, B)$, this is:

$$\frac{|A \cap C|}{|A \cup C|} + \frac{|C \cap B|}{|C \cup B|} \leq \frac{|A \cap B|}{|A \cup B|} + 1.$$

In the fig in [a4](#), I represent each part of intersection. The inequality above can be written as



$$\frac{e+g}{a+c+d+e+f+g} + \frac{f+g}{b+c+d+e+f+g} - \frac{d+g}{a+b+d+e+f+g} - 1 \leq 0.$$

Use simply to simplify the LHS to get $(-a^2b - a^2c - a^2d - a^2e - ab^2 - 2abc - 4abd - 2abe - 2abf - 2abg - ac^2 - 4acd - 2ace - 2acf - 2acg - 3ad^2 - 4ade - 3adf - 3adg - ae^2 - aef - aeg - b^2c - b^2d - b^2f - bc^2 - 4bcd - 2bce - 2bcf - 2bcg - 3bd^2 - 3bde - 4bdf - 3bdg - bef - bf^2 - bfg - 2c^2d - c^2e - c^2f - 2c^2g - 4cd^2 - 5cde - 5cdf - 6cdg - ce^2 - 2cef - 3ceg - cf^2 - 3cfg - 2cg^2 - 2d^3 - 4d^2e - 4d^2f - 4d^2g - 2de^2 - 4def - 4deg - 2df^2 - 4dfg - 2dg^2)/D$, where D is $\frac{1}{(a+c+d+e+f+g)(b+c+d+e+f+g)(a+b+d+e+f+g)}$, and we are really lucky to find the numerator already non-positive.

The Jaccard distance is a metric.

b. $\cos(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}$, simplified from the lecture PPT.

1) Anyway, cosine distance is a cosine value within $[-1, 1]$, which is not non-negative.

Cosine distance is not a metric.

c.

$$\text{levenshtein}(a, b) = \begin{cases} \max(i, j) & \text{if } \min(i, j) = 0, \\ \min \begin{cases} \text{levenshtein}(a[1..i-1], b[1..j]) + 1 \\ \text{levenshtein}(a[1..i], b[1..j-1]) + 1 \\ \text{levenshtein}(a[1..i-1], b[1..j-1]) + 1_{(a_i \neq b_j)} \end{cases} & \text{otherwise.} \end{cases}$$

It is hard to prove with pure equations. I'd use logical explanations.

- 1) Obviously, according to definition, you can never edit a string negative times.
- 2) A string dose not need to be edited to get itself.
- 3) You can inverse all the operations from string a to b to get a from b . If in either direction the operation count is minimal, in the other direction it must also be minimal, i.e. is a levenshtein distance.
- 4) Proof by contradiction: If you can edit a to c and then to b with less edit count than directly from a to b , the route you found would replace the levenshtein distance path from a to b so this would never happen.

The Levenshtein distance is a metric.

d. $H(\mathbf{a}, \mathbf{b}) = \sum_{i=1}^n (a_i \neq b_i)$

- 1) A sum of bool is no way negative.
- 2) The Hamming distance of a string with itself is always the 0 for it is the same as itself.
- 3) $(a_i \neq b_i)$ indicates $(b_i \neq a_i)$, which shows $H(\mathbf{a}, \mathbf{b}) = H(\mathbf{b}, \mathbf{a})$.
- 4) Here we derive a stronger conclusion:

To prove $H(\mathbf{a}, \mathbf{b}) \leq H(\mathbf{a}, \mathbf{c}) + H(\mathbf{c}, \mathbf{b})$, we'd prove

$$\forall i \in [1, n], 1(a_i \neq b_i) \leq 1(a_i \neq c_i) + 1(c_i \neq b_i) :$$

- $a_i = b_i$, we have $1(a_i \neq c_i) + 1(c_i \neq b_i) \geq 0 = 1(a_i \neq b_i)$.
- $a_i \neq b_i$, c_i can be equal to at most one of a_i and b_i , which is,

$$2 \geq 1(a_i \neq c_i) + 1(c_i \neq b_i) \geq 1 = 1(a_i \neq b_i)$$

The Hamming distance is a metric.

Where is the lemma used? Confused.

Question 2

1. Consider a uniform distribution inside a $L \times L$ square, the x and y coordinates of the position p denote the pair of the points in this problem. We are to calculate the expectation of $|x - y|$ inside the square:

$$\begin{aligned}
 \mathbb{E}[|x - y|] &= \int_0^L \int_0^L \frac{|x - y|}{L^2} dx dy \\
 &= 2 * \int_0^L \int_0^x \frac{x - y}{L^2} dx dy \\
 &= 2 * \int_0^L \frac{x^2}{2L^2} dx \\
 &= 2 * \frac{L^3}{6L^2} = \frac{L}{3}
 \end{aligned}$$

2. $\mathbb{E}[|x - y|^2] = \mathbb{E}[\sum_1^d |x_i - y_i|^2]$, and since the coordinates distributed uniformly in all dimensions,

$$\begin{aligned}
 \mathbb{E}[|x_i - y_i|^2] &= \sum_1^d \mathbb{E}[|x_i - y_i|^2] \\
 &= d \cdot \int_0^L \int_0^L \frac{|x - y|^2}{L^2} dx dy \\
 &= \frac{L^2 d}{6}
 \end{aligned}$$

3. The possible range of the minimum distance M is $[0, \frac{L}{N-1}]$. It's not easy to directly derive the probability of M . The distribution of all the points lie in an N dimensional cube with volume L^N . It's easier to imagine the case when M is larger than a certain m . Let x_n be sorted, i.e. $x_1 \leq x_2 \leq \dots \leq x_n$. This adds a probability scale $A_N^N = N!$. $P(M > m)$ shows $x_i + m < x_{i+1}, \forall i \in [1, N - 1]$. This operation causes the high-dimensional cube to collapse, reducing m on each axis. The length of the cube become $L - (N - 1)m$. Note that this collapse does not require the points to be sorted. To align the probability, we need to scale it by $\frac{1}{N!}$.

The probability $P(M > m)$ is thus given by $\frac{(L - (N - 1)m)^N}{L^N}$.

The expectation can be calculated by:

$$\begin{aligned}
 P(M) &= \int_0^{\frac{L}{N-1}} P(M > m) dm \\
 &= \int_0^{\frac{L}{N-1}} \frac{(L - (N - 1)m)^N}{L^N} dm \\
 &= -\frac{1}{N - 1} \cdot \frac{1}{L^N} \cdot \frac{(L - (N - 1)m)^{N+1}}{N + 1} \Big|_0^{\frac{L}{N-1}} \\
 &= \frac{L}{N^2 - 1}
 \end{aligned}$$

4. I don't really understand what the problem is asking. But since there is a minimum distance that is relatively low when N is large, it is efficient to use $KD - tree$ with a certain neighbour range based on $P(M)$ when clustering.

Original Questions

1. Please prove or disprove that the following distances are metrics.

- Jaccard distance
- cosine distance
- edit distance
- hamming distance

Hint: you may need to prove the lemma below at first:

For any set X and its subsets A, B, C , i.e. $A, B, C \subseteq X$, it holds that

$$|A \cap C| \cdot |B \cup C| + |A \cup C| \cdot |B \cap C| \leq |C| \cdot (|A| + |B|)$$

Figure 1: Question 1

2.

- Prove the **average** distance $\langle \mathbb{E}[|x - y|] \rangle$ between a pair of points $x, y \in [0, L]$ is $\frac{L}{3}$. (points x, y are uniformly distributed)
- Solve the **average** squared distance $\langle \mathbb{E}[\|x - y\|_2^2] \rangle$ for $x, y \in [0, L]^d$. (x, y are uniformly distributed and the result depend on L, d)
- If there are N points $\{x_n\} \in [0, L]$, prove that the **expectation** of **minimum** distances $\langle \mathbb{E}[\min |x - y|], x, y \in [0, L] \rangle$ between any two points is $\frac{L}{N^2 - 1}$.
- (Optional, no bonus, open question) Consider the clustering algorithm (k-means, for example) and what can we learn from the conclusions above?

Figure 2: Question 2

Reference

[stackexchange](#)

Finished other parts independently with help of GPT only with the tikz image.